

We will cover the following six topics in the Week 9 lectures:

- 1 Line integrals of plane scalar fields of the first and second kinds.
- 2 Line integrals of space scalar fields of the first, second, and third kinds.
- 3 Work done by a force field.
- 4 The Fundamental Theorem of Line Integrals.
- 5 Conservative vector fields.
- 6 Green's Theorem.

$\int_C f \, ds$ $\xrightarrow{\text{scalar field}}$: does not depend on orientation

$\int_C \vec{F} \cdot d\vec{r}$: depends on orientation
 $\downarrow$
vector field
reverse the direction of travel, then
you obtain the negative of the actual value

Line Integrals of Plane Scalar Fields — the First and Second Kinds

Let f be a plane scalar field, i.e., a scalar function of two real variables.

Let C be an oriented curve in $\mathbb{R}^2$ contained within $\text{Dom}(f)$.

Let $\mathbf{r} = (X, Y)$ be a nice parametrization of C that respects the orientation of C .

Suppose that the parameter space of $\mathbf{r}$ is $I = [a, b]$.

components
of $\vec{v}$

→ closed & bounded ~



Line Integrals of Plane Scalar Fields — the First and Second Kinds

If there exist $L_1, L_2 \in \mathbb{R}$ such that, regardless of the sampling t^* of I chosen,

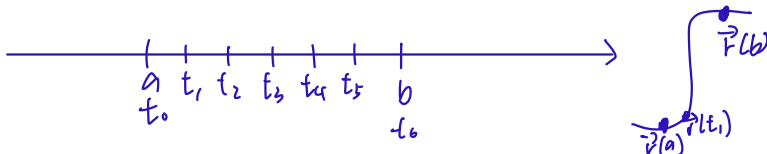
$$L_1 = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\mathbf{r}(t_{n,i}^*)) \cdot \underbrace{\left[X\left(a + i \left(\frac{b-a}{n}\right)\right) - X\left(a + (i-1) \left(\frac{b-a}{n}\right)\right) \right]}_{(\Delta x)_i};$$

Handwritten notes: x -value of t_{i-1} (above $a + (i-1)(\frac{b-a}{n})$) and x -value of t_i (above $a + i(\frac{b-a}{n})$)

$$L_2 = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\mathbf{r}(t_{n,i}^*)) \cdot \underbrace{\left[Y\left(a + i \left(\frac{b-a}{n}\right)\right) - Y\left(a + (i-1) \left(\frac{b-a}{n}\right)\right) \right]}_{(\Delta y)_i},$$

then

- denote L_1 by $\int_C f \, dx$ and call it the “line integral of f of the 1st kind along C ”;
- denote L_2 by $\int_C f \, dy$ and call it the “line integral of f of the 2nd kind along C ”.



Line Integrals of Plane Scalar Fields — the First and Second Kinds

Theorem (Sufficient Conditions for Existence)

- Let f be a continuous plane scalar field.
- Let C be a curve in $\mathbb{R}^2$, contained with $\text{Dom}(f)$, whose length exists.

Then $\int_C f \, dx$ and $\int_C f \, dy$ both exist.



Theorem (Continuous Plane Scalar Fields and Smooth Oriented Curves)

- Let f be a continuous plane scalar field.
- Let C be a smooth oriented curve in $\mathbb{R}^2$ contained within $\text{Dom}(f)$.

Then, for every orientation-respecting, smooth, nice parametrization $\mathbf{r} = (X, Y)$ of C , with parameter space $I = [a, b]$,

$$\left. \begin{aligned} \int_C f \, dx &= \int_a^b f(X(t), Y(t)) \cdot X'(t) \, dt, \\ \int_C f \, dy &= \int_a^b f(X(t), Y(t)) \cdot Y'(t) \, dt. \end{aligned} \right\} \quad \text{✱}$$

Line Integrals of Plane Scalar Fields — the First and Second Kinds

Let P and Q be plane scalar fields.

Let C be an oriented curve contained within $\text{Dom}(P) \cap \text{Dom}(Q)$.

Then it is very common to adopt the following notation:

$$\int_C (P \, dx + Q \, dy) \stackrel{\text{df}}{=} \int_C P \, dx + \int_C Q \, dy.$$

Theorem (Relation to Line Integrals of Plane Vector Fields)

- Let $\mathbf{F} = (P, Q)$ be a plane vector field.
- Let C be an oriented curve contained within $\text{Dom}(\mathbf{F})$.

If $\int_C P \, dx$ and $\int_C Q \, dy$ both exist, then $\int_C \mathbf{F} \bullet d\mathbf{r}$ exists, and

$$\int_C \mathbf{F} \bullet d\mathbf{r} = \int_C P \, dx + \int_C Q \, dy = \int_C (P \, dx + Q \, dy).$$

Handwritten notes:
→ First component, first field
↓ Second component, second time

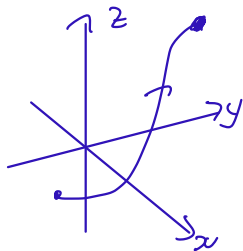
Line Integrals of Space Scalar Fields — the First, Second, and Third Kinds

Let f be a space scalar field, i.e., a scalar function of three real variables.

Let C be an oriented curve in $\mathbb{R}^3$ contained within $\text{Dom}(f)$.

Let $\mathbf{r} = (X, Y, Z)$ be a nice parametrization of C that respects the orientation of C .
components of $\mathbf{r}$

Suppose that the parameter space of $\mathbf{r}$ is $I = [a, b]$.



Line Integrals of Space Scalar Fields — the First, Second, and Third Kinds

If there exist $L_1, L_2, L_3 \in \mathbb{R}$ such that, regardless of the sampling t^* of I chosen,

$$L_1 = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\mathbf{r}(t_{n,i}^*)) \cdot \underbrace{\left[X\left(a + i\left(\frac{b-a}{n}\right)\right) - X\left(a + (i-1)\left(\frac{b-a}{n}\right)\right) \right]}_{(\Delta x)_i};$$

$$L_2 = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\mathbf{r}(t_{n,i}^*)) \cdot \underbrace{\left[Y\left(a + i\left(\frac{b-a}{n}\right)\right) - Y\left(a + (i-1)\left(\frac{b-a}{n}\right)\right) \right]}_{(\Delta y)_i};$$

$$L_3 = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(\mathbf{r}(t_{n,i}^*)) \cdot \underbrace{\left[Z\left(a + i\left(\frac{b-a}{n}\right)\right) - Z\left(a + (i-1)\left(\frac{b-a}{n}\right)\right) \right]}_{(\Delta z)_i},$$

then

- denote L_1 by $\int_C f \, dx$ and call it the “line integral of f of the 1st kind along C ”;
- denote L_2 by $\int_C f \, dy$ and call it the “line integral of f of the 2nd kind along C ”;
- denote L_3 by $\int_C f \, dz$ and call it the “line integral of f of the 3rd kind along C ”.

Theorem (Sufficient Conditions for Existence)

- Let f be a continuous space scalar field.
- Let C be a curve in $\mathbb{R}^3$, contained with $\text{Dom}(f)$, whose length exists.

Then $\int_C f \, dx$, $\int_C f \, dy$, and $\int_C f \, dz$ all exist.

Line Integrals of Space Scalar Fields — the First, Second, and Third Kinds

$\rightarrow X', Y', Z'$ will be defined.

Theorem (Continuous Space Scalar Fields and Smooth Oriented Curves)

- Let f be a continuous space scalar field.
- Let C be a smooth oriented curve in $\mathbb{R}^3$ contained within $\text{Dom}(f)$.

Then, for every orientation-respecting, smooth, nice parametrization $\mathbf{r} = (X, Y, Z)$ of C , with parameter space $I = [a, b]$,

$$\left. \begin{aligned} \int_C f \, dx &= \int_a^b f(X(t), Y(t), Z(t)) \cdot \boxed{X'(t)} \, dt, \\ \int_C f \, dy &= \int_a^b f(X(t), Y(t), Z(t)) \cdot \boxed{Y'(t)} \, dt, \\ \int_C f \, dz &= \int_a^b f(X(t), Y(t), Z(t)) \cdot \boxed{Z'(t)} \, dt. \end{aligned} \right\} \quad \otimes$$

Line Integrals of Space Scalar Fields — the First, Second, and Third Kinds

Let P , Q , and R be space scalar fields.

Let C be an oriented curve contained within $\text{Dom}(P) \cap \text{Dom}(Q) \cap \text{Dom}(R)$.

Then it is very common to adopt the following notation:

$$\int_C (P \, dx + Q \, dy + R \, dz) \stackrel{\text{df}}{=} \int_C P \, dx + \int_C Q \, dy + \int_C R \, dz.$$

Theorem (Relation to Line Integrals of Space Vector Fields)

- Let $\mathbf{F} = (P, Q, R)$ be a space vector field.
- Let C be an oriented curve contained within $\text{Dom}(\mathbf{F})$.

If $\int_C P \, dx$, $\int_C Q \, dy$, and $\int_C R \, dz$ all exist, then $\int_C \mathbf{F} \bullet d\mathbf{r}$ exists, and

$$\int_C \mathbf{F} \bullet d\mathbf{r} = \int_C P \, dx + \int_C Q \, dy + \int_C R \, dz = \int_C (P \, dx + Q \, dy + R \, dz).$$

Directed Line Segments

A **directed line segment** is a line segment with a direction of travel.

We usually specify the direction of a directed line segment using its endpoints.

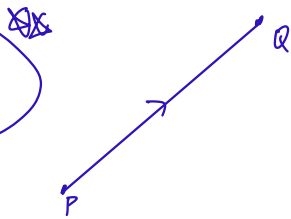
Let $n \in \mathbb{N}$, and let L be a directed line segment in $\mathbb{R}^n$ going from an endpoint P to an endpoint Q .

Then the vector function $\mathbf{r} : [0, 1] \rightarrow \mathbb{R}^n$ defined by

$$\mathbf{r}(t) \stackrel{\text{df}}{=} \overrightarrow{OP} + t \cdot \overrightarrow{PQ}$$

is an orientation-respecting parametrization of L .

We call this parametrization of L the “**standard affine parametrization of L** ”.



$$\overrightarrow{OQ} - \overrightarrow{OP}$$

Directed Line Segments — Example 1

Example

Let L denote the directed line segment in $\mathbb{R}^2$ going from $(0, 2)$ to $(1, 4)$.

Problem: Evaluate the line integral $\int_L [x^2 y \, dx + \sin(\pi y) \, dy]$.

Solution $\vec{r} : [0, 1] \rightarrow \mathbb{R}^2$, $\vec{r}(t) = (0, 2) + t((1, 4) - (0, 2))$
 $= (0, 2) + t(1, 2)$
 $= (t, 2t+2)$
 $\downarrow \quad \downarrow$
 $x(t) \quad y(t)$

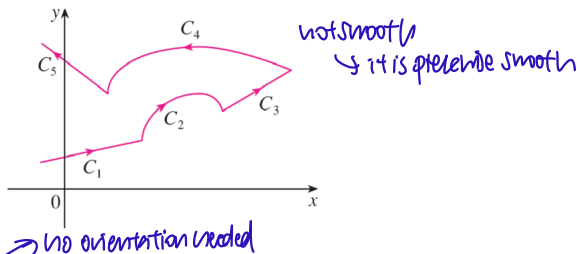
$$x'(t) = 1, \quad y'(t) = 2$$

$$\begin{aligned} \int_L x^2 y \, dx &= \int_0^1 t^2 (2t+2) \cdot 1 \, dt \\ &= \int_0^1 (2t^3 + 2t^2) \, dt \\ &= \left[\frac{t^4}{2} + \frac{2t^3}{3} \right]_0^1 = \frac{7}{6} \end{aligned}$$

$$\begin{aligned} \int_L \sin(\pi y) \, dy &= \int_0^1 \sin(\pi(2t+2)) \cdot 2 \, dt \\ &= \int_0^1 2 \sin(2\pi(t+1)) \, dt \\ &= \left[-\frac{2 \cos(2\pi(t+1))}{2\pi} \right]_0^1 \\ &= \left(-\frac{1}{\pi}\right) - \left(-\frac{1}{\pi}\right) = 0 \end{aligned}$$

Ans: $\frac{7}{6} + 0 = \frac{7}{6}$

Additivity of Line Integrals of Scalar Fields over Sub-Curves



Theorem (Additivity of Line Integrals of Scalar Fields over Sub-Curves)

- Let f be a scalar field of several real variables.
- Let C be a curve contained within $\text{Dom}(f)$.
- Suppose that C is comprised of k curves, $C_1, \dots, C_k$, joined end-to-end.

If $\int_{C_1} f \, ds, \dots, \int_{C_k} f \, ds$ all exist, then $\int_C f \, ds$ exists, and

$$\int_C f \, ds = \int_{C_1} f \, ds + \dots + \int_{C_k} f \, ds.$$

Additivity of Line Integrals of Vector Fields over Oriented Sub-Curves

must have orientation

Theorem (Additivity of Line Integrals of Vector Fields over Oriented Sub-Curves)

- Let $n \in \mathbb{N}$, and let $\mathbf{F}$ be an n -dimensional vector field.
- Let C be an oriented curve in $\mathbb{R}^n$ contained within $\text{Dom}(\mathbf{F})$.
- Suppose that C is comprised of k oriented curves, $C_1, \dots, C_k$, joined end-to-end.

If $\int_{C_1} \mathbf{F} \bullet d\mathbf{r}, \dots, \int_{C_k} \mathbf{F} \bullet d\mathbf{r}$ all exist, then $\int_C \mathbf{F} \bullet d\mathbf{r}$ exists, and

$$\int_C \mathbf{F} \bullet d\mathbf{r} = \int_{C_1} \mathbf{F} \bullet d\mathbf{r} + \dots + \int_{C_n} \mathbf{F} \bullet d\mathbf{r}.$$

Piecewise-Smooth Curves

Very often, we will find ourselves working with curves (or oriented curves) that are not smooth on the overall but still smooth on sub-curves (or oriented sub-curves).

Definition (Piecewise-Smooth Curves (or Oriented Curves))

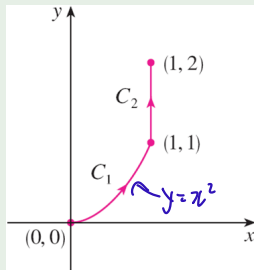
A curve (or oriented curve) is said to be **piecewise-smooth** if and only if it is comprised of finitely many curves (or oriented curves) joined end-to-end, each of which is smooth.

Additivity of the Line Integral over Sub-Curves — Example 1

Example

Let C denote the piecewise-smooth curve in $\mathbb{R}^2$ comprised of the following sub-curves:

- The arc C_1 of the parabola given by the equation $y = x^2$ from $(0, 0)$ to $(1, 1)$.
- The vertical line segment C_2 from $(1, 1)$ to $(1, 2)$.



Problem: Evaluate the line integral $\int_C 2x \, ds$.

orientation is not imp here

Additivity of the Line Integral over Sub-Curves — Example 1 (cont'd)

Solution

$$\textcircled{1} \quad \vec{r}_1: [0, 1] \rightarrow \mathbb{R}^2, \quad \vec{r}_1(t) = (t, t^2) \quad || \vec{r}_1'(t) = (1, 2t)$$

$$\textcircled{2} \quad \vec{r}_2: [1, 2] \rightarrow \mathbb{R}^2, \quad \vec{r}_2(t) = (1, t) \quad || \vec{r}_2'(t) = (0, 1)$$

$$\begin{aligned} \int_{C_1} 2x \, ds &= \int_0^1 2t \cdot ||\vec{r}_1'(t)|| \, dt = \int_0^1 2t \cdot \sqrt{1+4t^2} \, dt \\ &= \left[\frac{1}{6} \sqrt{(1+4t^2)^3} \right]_0^1 \\ &= \frac{1}{6} (5\sqrt{5} - 1) \end{aligned}$$

$$\begin{aligned} \int_{C_2} 2x \, ds &= \int_1^2 2 ||\vec{r}_2'(t)|| \, dt \\ &= \int_1^2 2 \cdot 1 \, dt \\ &= \int_1^2 2 \, dt \\ &= [2t]_1^2 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \int_C 2x \, ds &= \int_{C_1} 2x \, ds + \int_{C_2} 2x \, ds \\ &= \frac{1}{6} (5\sqrt{5} - 1) + 2 \end{aligned}$$

Work Done by a Force Field — Force Fields

A plane (or space) vector field can be visualized as a force field in $\mathbb{R}^2$ (or $\mathbb{R}^3$).

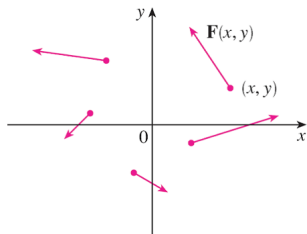


Figure: A plane vector field.

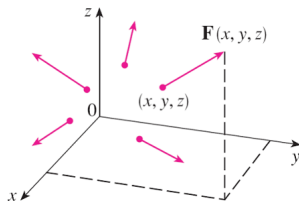


Figure: A space vector field.

Example (Examples of force fields)

- Gravitational fields.
- Electrical fields (usually denoted by $\mathbf{E}$).
- Magnetic fields (usually denoted by $\mathbf{B}$).

Work Done by Force Field — Example 1

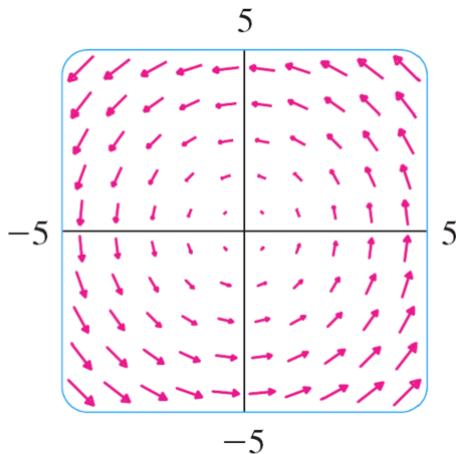


Figure: The plane vector field $\mathbf{F}$ on $\mathbb{R}^2$, defined by $\mathbf{F}(x, y) \stackrel{\text{df}}{=} -y \cdot \mathbf{i} + x \cdot \mathbf{j}$. $(-y, x)$

Work Done by Force Field — Example 2

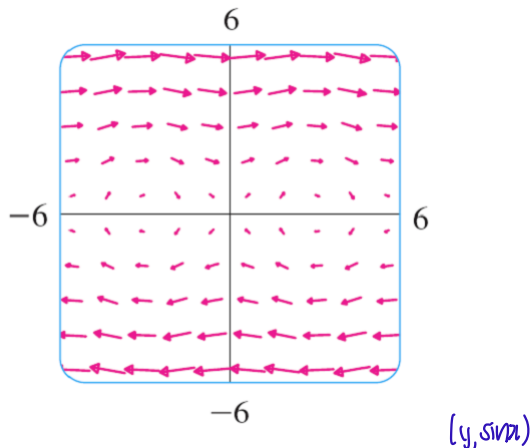


Figure: The plane vector field $\mathbf{F}$ on $\mathbb{R}^2$, defined by $\mathbf{F}(x, y) \stackrel{\text{df}}{=} y \cdot \mathbf{i} + \sin(x) \cdot \mathbf{j}$.

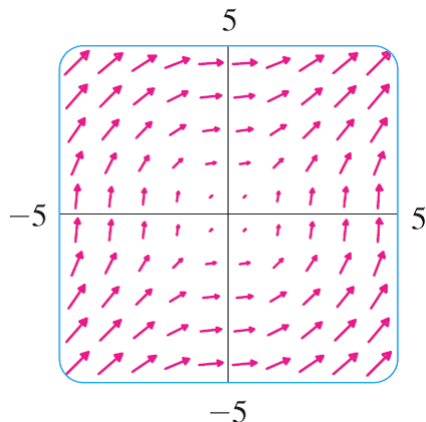
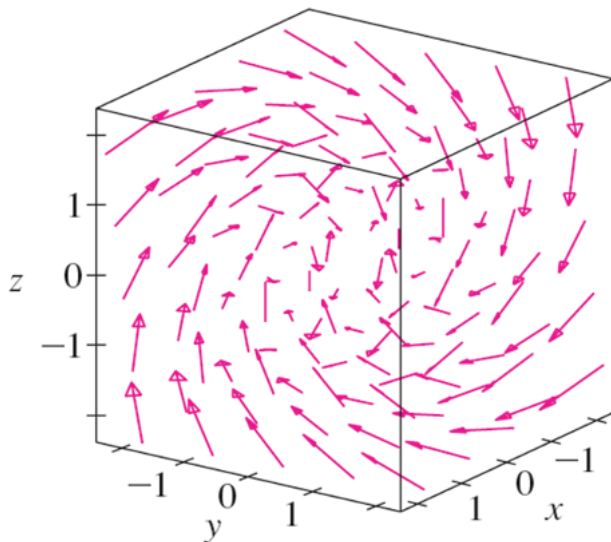


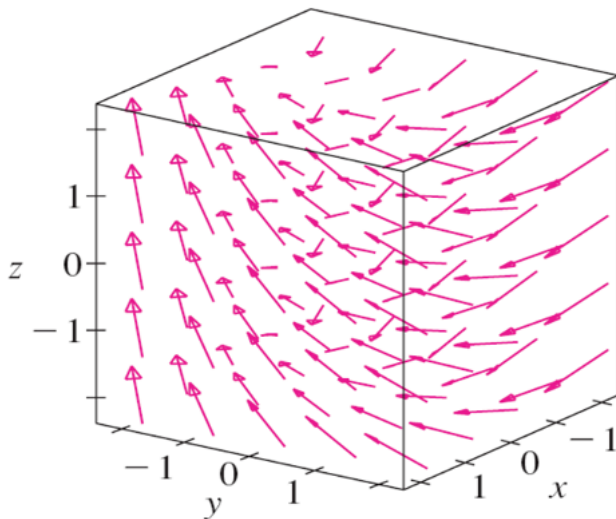
Figure: The plane vector field $\mathbf{F}$ on $\mathbb{R}^2$, defined by $\mathbf{F}(x, y) \stackrel{\text{df}}{=} \ln(1 + y^2) \cdot \mathbf{i} + \ln(1 + x^2) \cdot \mathbf{j}$.

Work Done by Force Field — Example 4



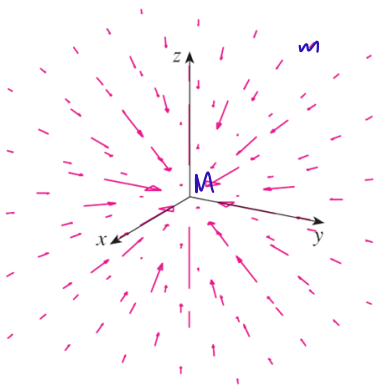
$$\mathbf{F}(x, y, z) = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$$

Work Done by Force Field — Example 5



$$\mathbf{F}(x, y, z) = y\mathbf{i} - 2\mathbf{j} + x\mathbf{k}$$

Work Done by Force Field — Example 6



these are just examples,
no need to know the
formulas!

The gravitational field $\mathbf{F}$ acting on a point mass m located in $\mathbb{R}^3 \setminus \{(0, 0, 0)\}$ generated by a point mass M located at the origin, given by

$$\mathbf{F}(x, y, z) = -\frac{GMm}{\|(x, y, z)\|^3} \cdot (x, y, z).$$

↓
cos the field points inwards

Recall from high-school physics that when a point object travels in a straight line from a point A to a point B while acted upon by a constant force $\mathbf{F}$, the work W done by $\mathbf{F}$ (and only $\mathbf{F}$) is given by

$$W = \mathbf{F} \bullet \overrightarrow{AB}.$$

Remark: The point object may be acted upon by other forces at the same time.

Work Done by Force Field — Work Done by a Force Along a Curve

The formula on the preceding slide is no longer valid in the following two scenarios:

- The point object travels along a straight line, but $\mathbf{F}$ is not constant along it.
- The point object does not travel along a straight line.

Definition (Work Done by a Force Field Along an Oriented Curve)

- Let $n \in \mathbb{N}$, and let $\mathbf{F}$ be an n -dimensional force field.
- Let C be an oriented curve in $\mathbb{R}^n$ contained within $\text{Dom}(\mathbf{F})$.

Then the **work** W done by $\mathbf{F}$ (and only $\mathbf{F}$) in moving a point object along C is defined as follows:

$$W \stackrel{\text{df}}{=} \int_C \mathbf{F} \bullet d\mathbf{r}. \quad \text{★ ★ ★}$$

This illustrates why the concept of line integrals is so important in physics.

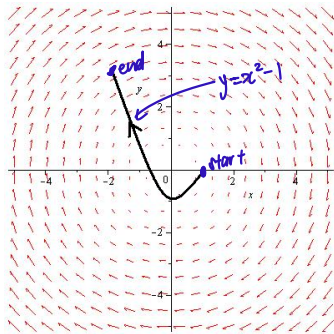
Work Done by Force Field — Example 1

Example

- A plane force field $\mathbf{F}$ on $\mathbb{R}^2$ is given by $\mathbf{F}(x, y) = (y, -x)$.
- A point object in $\mathbb{R}^2$ is acted upon by $\mathbf{F}$ while moving along the parabola given by the equation $y = x^2 - 1$ from $(\underline{1}, 0)$ to $(\underline{-2}, 3)$.

Problem: Find the work done by $\mathbf{F}$ in moving the point object along its path.

Solution



Work Done by Force Field — Example 1 (cont'd)

Let C denote the oriented curve representing the point object's motion

Define a vector function $\vec{r}: [-2, 1] \rightarrow \mathbb{R}^2$ by $\vec{r}(t) = (t, t^2 - 1)$ *must write this way*

Observe that $\vec{r}$ parameterizes $-C$ instead of C .

$$\begin{aligned} W &= \int_C \vec{F} \cdot d\vec{r} = - \int_{-C} \vec{F} \cdot d\vec{r} \\ &= - \int_{-2}^1 \vec{F}(\vec{r}(t)) \cdot \vec{r}'(t) dt \\ &= - \int_{-2}^1 \vec{F}(t, t^2 - 1) \cdot (1, 2t) dt \\ &= - \int_{-2}^1 (t^2 - 1, -t) \cdot (1, 2t) dt \\ &= - \int_{-2}^1 t^2 - 1 - 2t^2 dt \\ &= - \int_{-2}^1 (-t^2 - 1) dt \\ &= \int_{-2}^1 (t^2 + 1) dt \\ &= \left[\frac{t^3}{3} + t \right]_{-2}^1 \\ &= \frac{4}{3} - \left(-\frac{14}{3} \right) \\ &= 6 \end{aligned}$$

Example

- A space force field $\mathbf{F}$ on $\mathbb{R}^3$ is given by $\mathbf{F}(x, y, z) = (y, x, z)$.
- A point object in $\mathbb{R}^3$ is acted upon by $\mathbf{F}$ while moving as follows:
 - Firstly, along the twisted cubic, given by

$$\{(x, x^2, x^3) \in \mathbb{R}^3 \mid x \in \mathbb{R}\}, \quad \left. \vphantom{\{(x, x^2, x^3) \in \mathbb{R}^3 \mid x \in \mathbb{R}\}} \right\} L_1$$

from $(0, 0, 0)$ to $(1, 1, 1)$.

- Secondly, along the line segment from $(1, 1, 1)$ to $(0, 1, 0)$. $\left. \vphantom{(1, 1, 1)} \right\} L_2$

Problem: Find the work done by $\mathbf{F}$ in moving the point object along its path.

Solution

$$L_1 \quad \vec{r}_1: [0, 1] \rightarrow \mathbb{R}^3, \quad \vec{r}_1(t) = (t, t^2, t^3), \quad \vec{r}_1'(t) = (1, 2t, 3t^2)$$

$$\begin{aligned} L_2 \quad \vec{r}_2: [0, 1] \rightarrow \mathbb{R}^3, \quad \vec{r}_2(t) &= \overbrace{(1, 1, 1)}^{\text{OP}} + t \cdot \overbrace{[(0, 1, 0) - (1, 1, 1)]}^{\text{PO}} \\ &= (1, 1, 1) + t(-1, 0, -1) \\ &= (-t+1, 1, -t+1) \end{aligned}$$

$$\vec{r}_2'(t) = (-1, 0, -1)$$

Work Done by Force Field — Example 2 (cont'd)

→ work done over C_1

$$\begin{aligned} W_1 &= \int_{C_1} \vec{F} \cdot d\vec{r} = \int_0^1 \vec{F}(\vec{r}_1(t)) \cdot \vec{r}_1'(t) dt \\ &= \int_0^1 \vec{F}(t, t^2, t^3) \cdot (1, 2t, 3t^2) dt \\ &= \int_0^1 (t^2, t, t^3) \cdot (1, 2t, 3t^2) dt \\ &= \int_0^1 (3t^2 + 3t^5) dt \\ &= 3 \int_0^1 (t^2 + t^5) dt \\ &= 3 \left[\frac{t^3}{3} + \frac{t^6}{6} \right]_0^1 = \frac{5}{2} \end{aligned}$$

$$\begin{aligned} W_2 &= \int_{C_2} \vec{F} \cdot d\vec{r} = \int_0^1 \vec{F}(\vec{r}_2(t)) \cdot \vec{r}_2'(t) dt \\ &= \int_0^1 \vec{F}(-t+1, 1, -t+1) \cdot (-1, 0, -1) dt \\ &= \int_0^1 (1, -t+1, -t+1) \cdot (-1, 0, -1) dt \\ &= \int_0^1 (t-2) dt \\ &= \left[\frac{t^2}{2} - 2t \right]_0^1 = -\frac{3}{2} \end{aligned}$$

$$W = W_1 + W_2 = \frac{5}{2} + \left(-\frac{3}{2}\right) = 1$$

The Fundamental Theorem of Line Integrals

You should be able to tell by now that it can be quite tedious to compute $\int_C \mathbf{F} \bullet d\mathbf{r}$.

However, ...

If $\mathbf{F}$ is a gradient vector field, i.e, there exists a scalar field f on $\text{Dom}(\mathbf{F})$ such that

$$\mathbf{F} = \nabla f,$$

then computing $\int_C \mathbf{F} \bullet d\mathbf{r}$ is super easy!



Theorem (The Fundamental Theorem of Line Integrals (FTLI))

- Let f be a real-valued function of several real variables, with the property that ∇f is continuous throughout $\text{Dom}(\nabla f)$.
- Let C be a piecewise-smooth oriented curve, contained within $\text{Dom}(\nabla f)$, that has initial point A and final point B (**note**: A and B are points on the curve).

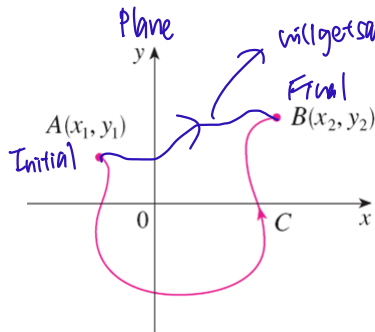
Then $\int_C \nabla f \bullet d\mathbf{r} = f(B) - f(A)$.

Remark: Does this not remind you of the Fundamental Theorem of Calculus?

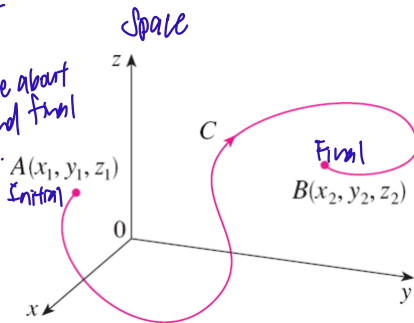
$\hookrightarrow \int_a^b f'(x) dx = f(b) - f(a)$



The Fundamental Theorem of Line Integrals



will get same answer
↓
only care about
initial and final
points.



$$\int_C \nabla f \bullet d\mathbf{r} = f(B) - f(A), \text{ no matter what } C \text{ (piecewise-smooth and oriented) is!}$$



Definition (Conservative Vector Fields)

Let $n \in \mathbb{N}$, and let $\mathbf{F}$ be an n -dimensional vector field.

Then $\mathbf{F}$ is said to be **conservative** if and only if there exists a real-valued function f of n real variables such that $\mathbf{F} = \nabla f$, in which case we call f a "**potential function of $\mathbf{F}$** ".

is the gradient of f

Remark: If $\mathbf{F}$ is conservative, then its domain is necessarily an open subset of $\mathbb{R}^n$.

Conservative Vector Fields — Example 1

Example

Define a plane vector field $\mathbf{F}$ on $\mathbb{R}^2$ by $\mathbf{F}(x, y) \stackrel{\text{df}}{=} (2xy, 1 + x^2 - y^2)$.

Problem: Show that $\mathbf{F}$ is conservative by finding a potential function of $\mathbf{F}$.

Solution

Find a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $\vec{F} = \nabla f = (f_x, f_y)$

For all $x, y \in \mathbb{R}$: ① $f_x(x, y) = 2xy$

$$\textcircled{2} f_y(x, y) = 1 + x^2 - y^2$$

not vly a time constant,
may depend on y !

$$\cdot \int 2xy \, dx = \textcircled{x^2 y} + \boxed{c(y)} \quad \text{would like this to be } f(x, y)$$

may depend on x

$$\cdot \int 1 + x^2 - y^2 \, dy = \boxed{y} + \textcircled{x^2 y} - \boxed{\frac{1}{3}y^3} + \underbrace{d(x)}_{=0} \Rightarrow \text{also like this to be } f(x, y)$$

• I can choose f so that

$$\forall x, y \in \mathbb{R}: f(x, y) = x^2 y + y - \frac{y^3}{3} \Rightarrow \text{check}$$

$$\text{can also choose } f(x, y) = x^2 y + y - \frac{y^3}{3} + 1$$

How to determine whether or not a given continuous vector field $\mathbf{F}$ is conservative

- ① If $\text{Dom}(\mathbf{F})$ is not open, then $\mathbf{F}$ is not conservative.
- ② If $\text{Dom}(\mathbf{F})$ is open but the FTLI fails, then $\mathbf{F}$ is not conservative.

Remark: Failure of the FTLI simply means that we can find two piecewise-smooth oriented curves, C_1 and C_2 , contained within $\text{Dom}(F)$, with the same initial points and the same final points, such that

$$\int_{C_1} \mathbf{F} \bullet d\mathbf{r} \neq \int_{C_2} \mathbf{F} \bullet d\mathbf{r}.$$

- By defⁿ ③ If $\mathbf{F}$ has a potential function, then $\mathbf{F}$ is conservative.

Remark: Refer to the last example.

Theorem (Test for Conservativeness on an Open and Simply Connected Domain)

Let $\mathbf{F} = (P, Q)$ be a plane vector field with the following three properties:

- $\text{Dom}(\mathbf{F})$ is open.
- $\text{Dom}(\mathbf{F})$ is simply connected, i.e., the following two statements hold:
 - Every pair of points in $\text{Dom}(\mathbf{F})$ is connected by a curve contained within $\text{Dom}(\mathbf{F})$.
 - $\text{Dom}(\mathbf{F})$ has no “holes”. $\Rightarrow$ donut

Remark: In the jargon of algebraic topology, this means that the fundamental group of $\text{Dom}(\mathbf{F})$ is trivial.
- $\mathbf{F}$ is smooth, i.e., all four first-order partial derivatives of P and Q are defined and continuous throughout $\text{Dom}(\mathbf{F})$.

Then $\mathbf{F}$ is conservative if and only if $\begin{matrix} (\Rightarrow) \text{ easy to prove this direction} \\ \text{easier to remember} \end{matrix}$

$$\begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ P & Q \end{vmatrix} \equiv 0 \quad \text{or, equivalently,} \quad \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}.$$

We can prove the forward implication of the theorem rather easily:

- Suppose that $\mathbf{F}$ is conservative.
- Let f be a potential function for $\mathbf{F}$, so that

$$(P, Q) = \mathbf{F} = \nabla f = (f_x, f_y). \quad \left. \begin{array}{l} P = f_x \\ Q = f_y \end{array} \right\}$$

- Then $P_y = f_{xy}$ and $Q_x = f_{yx}$.
- As P_y and Q_x are continuous, Clairaut's Theorem therefore yields

$$P_y = f_{xy} = f_{yx} = Q_x.$$

- This completes the proof. ■

Conservative Vector Fields — Open and Simply Connected Domains

Theorem (Test for Conservativeness on an Open and Simply Connected Domain)

Let $\mathbf{F} = (P, Q, R)$ be a space vector field with the following three properties:

- $\text{Dom}(\mathbf{F})$ is open.
- $\text{Dom}(\mathbf{F})$ is simply connected, i.e., the following two statements hold:
 - Every pair of points in $\text{Dom}(\mathbf{F})$ is connected by a curve contained within $\text{Dom}(\mathbf{F})$.
 - $\text{Dom}(\mathbf{F})$ has no “holes”.

Remark: In the jargon of algebraic topology, this means that the fundamental group of $\text{Dom}(\mathbf{F})$ is trivial.
- $\mathbf{F}$ is smooth, i.e., all nine first-order partial derivatives of P , Q , and R are defined and continuous throughout $\text{Dom}(\mathbf{F})$.

Then $\mathbf{F}$ is conservative if and only if

$$\underbrace{\begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}}_{\text{curl } (\vec{F})} \equiv (0, 0, 0) \quad \text{or, equivalently,} \quad \begin{cases} \frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} \\ \frac{\partial R}{\partial x} = \frac{\partial P}{\partial z} \\ \frac{\partial R}{\partial y} = \frac{\partial Q}{\partial z} \end{cases}.$$

easier to remember

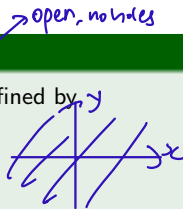
Conservative Vector Fields — Example 2

Example

Problem: Determine whether the plane vector field $\mathbf{F}$ on $\mathbb{R}^2$ defined by

$$\mathbf{F}(x, y) \stackrel{\text{df}}{=} \left(\underbrace{3 + 2xy}_{P(x, y)}, \underbrace{x^2 - 3y^2}_{Q(x, y)} \right)$$

is conservative.



$$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y} [3 + 2xy] = 2x$$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x} [x^2 - 3y^2] = 2x$$

$\therefore \mathbf{F}$ is conservative

Methods of Computing Line Integrals

We have learned two methods of computing line integrals of continuous vector fields:

- 1 Using a smooth, orientation-respecting, nice parametrization:

Direct Method

$$\int_C \mathbf{F} \bullet d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \bullet \mathbf{r}'(t) dt.$$

- 2 Using the Fundamental Theorem of Line Integrals:

$$\int_C \nabla f \bullet d\mathbf{r} = f(B) - f(A). \quad \text{no parametrization needed}$$

Remark: This assumes that the vector field is conservative.

We will learn two more methods:

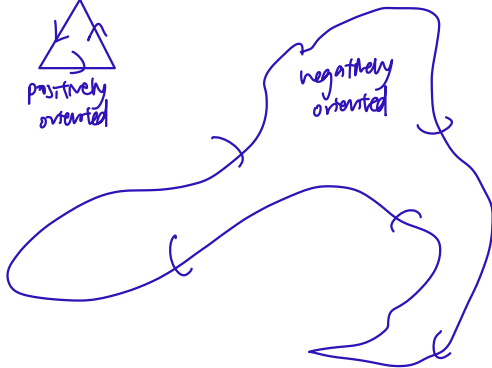
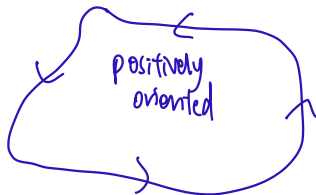
- 3 Green's Theorem (for plane vector fields).
- 4 Stokes' Theorem (for space vector fields).

A Special Symbol for the Line Integral over a Loop

Moving forward, we shall write " $\oint_C$ " for " $\int_C$ " whenever C is a **positively oriented loop**.

Remark: A positive orientation means a counterclockwise direction of travel.

The symbol " $\oint$ " is commonly used in complex analysis to denote contour integrals.



The Jordan Curve Theorem

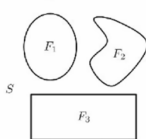
Theorem (The Jordan Curve Theorem)

The complement of a loop in the plane $\mathbb{R}^2$ is the union of two disjoint connected open regions, one bounded (the “interior”) and the other unbounded (the “exterior”).

This theorem is notoriously difficult to prove, but some people beg to differ ...

Proof of the Jordan Curve Theorem

The Jordan curve theorem states that a non-self-intersecting continuous loop in the plane (i.e. a Jordan curve) divides the plane into an interior region and an exterior region.



Jordan Curves



Not Jordan Curves

(Bangalore Ravi Kiran “Energetic-Lattice based optimization”, Figure 4)

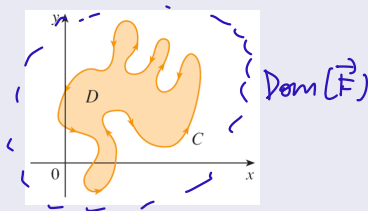
We shall use proof by f[REDACTED] obviousness. It's a closed loop. Of course there's going to be an outside and inside. Humans have been making borders and fences for millennia and this still needs to be proven?? What am I supposed to say? It's so bloody obvious!

It's like trying to prove $1+1=2$. Why is this even a theorem? Not even worth to be a Lemma or Corollary tbh. This trivial a[REDACTED] b[REDACTED]

And thus concludes the proof of the “theorem”.

Theorem (Green's Theorem)

- Let C be a positively oriented loop in $\mathbb{R}^2$ whose length exists.



- Let D denote the bounded and closed region in $\mathbb{R}^2$ made up of C and its interior.

Remark: The Jordan Curve Theorem guarantees that an interior exists at all.

- Let $\mathbf{F} = (P, Q)$ be a smooth plane vector field whose domain is an open subset of $\mathbb{R}^2$ containing D .
 components of F

Then $\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA(x, y)$ exists, and

$$\oint_C \mathbf{F} \bullet d\mathbf{r} = \oint_C (P \, dx + Q \, dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA(x, y).$$

Double Integral

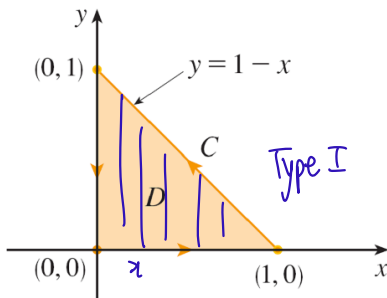
Example

Let C denote the positively oriented triangle in $\mathbb{R}^2$ with vertices

$$(0, 0), \quad (1, 0), \quad \text{and} \quad (0, 1).$$

Problem: Evaluate the line integral $\oint_C \left(x^4 \overset{P}{dx} + xy \overset{Q}{dy} \right)$.

Solution



Green's Theorem — Example 1 (cont'd)

Define $P, Q : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$P(x, y) = x^4 \text{ and } Q(x, y) = xy$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = y - 0 = y$$

$$\begin{aligned}\oint_C (x^4 dx + xy dy) &= \iint_D y \, dA(x, y) \\ &= \int_0^1 \left[\int_0^{1-x} y \, dy \right] dx \\ &= \int_0^1 \left[\frac{y^2}{2} \right]_{y=0}^{y=1-x} dx \\ &= \int_0^1 \frac{(1-x)^2}{2} dx \\ &= \left[-\frac{(1-x)^3}{6} \right]_0^1 \\ &= 0 - \left(-\frac{1}{6}\right) \\ &= \frac{1}{6}\end{aligned}$$

Example

Let C denote the positively oriented circle in $\mathbb{R}^2$ with center $(0,0)$ and radius 3.

Problem: Evaluate $\oint_C \left[\underbrace{(3y - e^{\sin(x)})}_{P(x,y)} dx + \underbrace{(7x + \sqrt{1+y^4})}_{Q(x,y)} dy \right]$.

Solution

Define $P, Q: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$P(x,y) = 3y - e^{\sin(x)}, \quad Q(x,y) = 7x + \sqrt{1+y^4}$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = 7 - 3 = 4$$

$$\begin{aligned} \oint_C \dots &= \iint_D 4 \, dA(x,y) \\ &= 4 \iint_D 1 \, dA(x,y) \\ &= 4 \text{Area}(D) \\ &= 4 \cdot \pi(3)^2 \\ &= 36\pi \end{aligned}$$

$$\text{Area} (?) = \iint_D 1 \, dA$$



Green's Theorem is often used to compute the area of the closed region in $\mathbb{R}^2$ bounded by a loop whose length exists.

Let C be a loop in $\mathbb{R}^2$ whose length exists.



Let D denote the bounded and closed region in $\mathbb{R}^2$ made up of C and its interior.

We wish to find $\text{Area}(D)$, which we know is equal to $\iint_D 1 \, dA(x, y)$ — if it exists.

If we can find functions $P, Q : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \equiv 1,$$

then Green's Theorem yields

$$\text{Area}(D) = \iint_D 1 \, dA(x, y) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA(x, y) = \underbrace{\oint_C (P \, dx + Q \, dy)}_{\text{line integral}}.$$

We have a few options for P and Q to choose from:

① $P \equiv 0$ and $Q(x, y) = x$.

② $P(x, y) = -y$ and $Q \equiv 0$.

③ $P(x, y) = -\frac{1}{2}y$ and $Q(x, y) = \frac{1}{2}x$. $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \frac{1}{2} - (-\frac{1}{2}) = 1$

Theorem (The Area Formula)

- Let C be a positively oriented loop in $\mathbb{R}^2$ whose length exists.
- Let D be the region in $\mathbb{R}^2$ made up of C and its interior.

Then $\text{Area}(D)$ exists, and

$$\text{Area}(D) = \oint_C x \, dy = - \oint_C y \, dx = \frac{1}{2} \oint_C (x \, dy - y \, dx).$$

→ 2D quantity

It is remarkable that for a loop in $\mathbb{R}^2$ whose length exists, the area of the closed region in $\mathbb{R}^2$ bounded by the loop can be calculated from the loop itself!

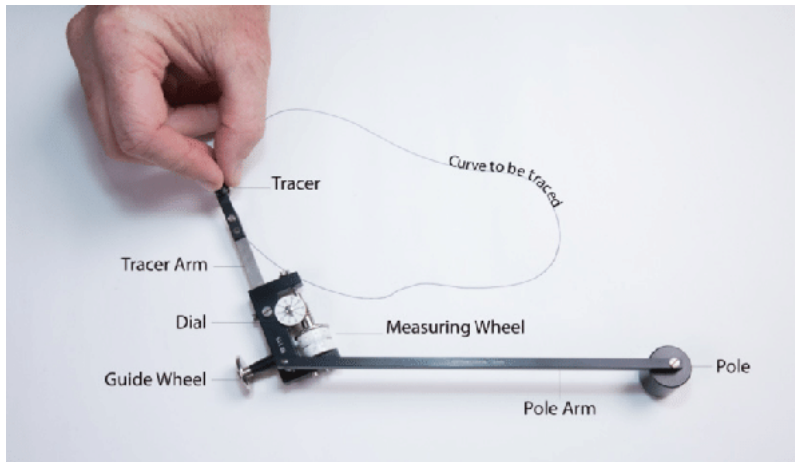


Figure: A planimeter (source: Eggers, *A Closer Look at the Compensating Polar Planimeter*)

This device, called a “planimeter”, applies Green's Theorem to compute plane areas!

Example

Let a and b be positive real numbers.

Problem: Find the area of the closed region in $\mathbb{R}^2$ bounded by the ellipse given by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Solution

Define a vector function $\vec{r}: [0, 1] \rightarrow \mathbb{R}^2$ by
 $\vec{r}(t) = (\underbrace{a \cos(t)}_{X(t)}, \underbrace{b \sin(t)}_{Y(t)})$

$\vec{r}$ gives us a parametrisation of the positively oriented ellipse

$$\begin{aligned} \text{Area}(D) &= \oint_C x \, dy = \int_0^{2\pi} \overset{\text{we are moving twice around!}}{X(t) \cdot Y'(t)} \, dt \\ &= \int_0^{2\pi} a \cos(t) \cdot b \cos(t) \, dt \\ &= ab \int_0^{2\pi} \cos^2(t) \, dt = ab \int_0^{2\pi} \left[\frac{1}{2} + \frac{1}{2} \cos(2t) \right] \, dt \\ &= ab \left[\frac{1}{2} t + \frac{1}{4} \sin(2t) \right]_0^{2\pi} = \pi ab \end{aligned}$$